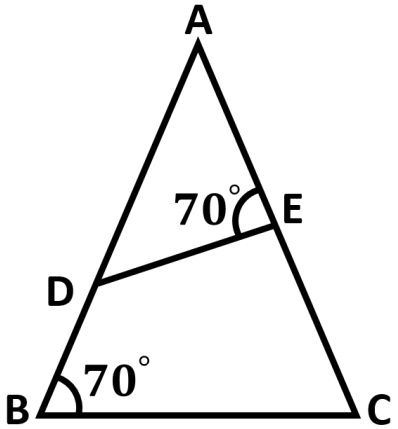
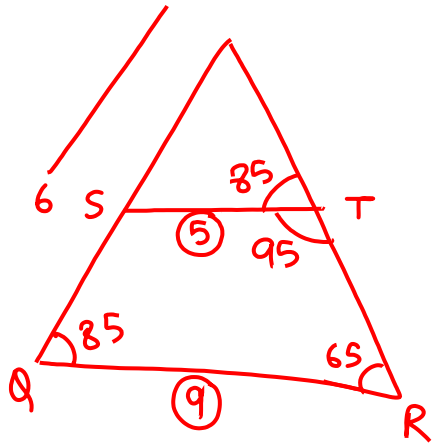


1. In the given figure, If $AD = 12cm, AE = 8cm$ and $EC = 14cm$, then what is the value in (in cm) of BD?

a) $\frac{50}{3}$ b) 15 c) $\frac{8}{3}$ d) $\frac{44}{3}$

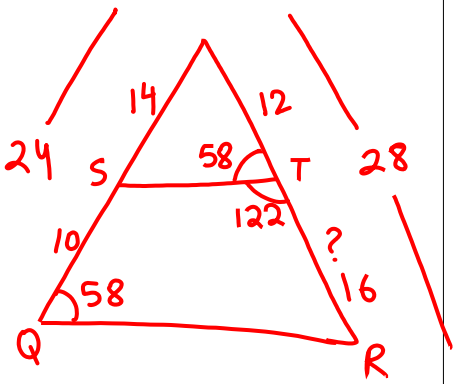
CGL 2017 T2



2. In $\Delta PQR, \angle Q = 85°$ and $\angle R = 65°$. Points S and T are on the sides PQ and PR respectively such that $\angle STR = 95°$ and the ratio of QR and ST is $9:5$. If $PQ = 21.6\ cm$, then the length of PT is:

a) 10.5 cm
b) 9.6 cm
c) 12 cm
d) 9 cm

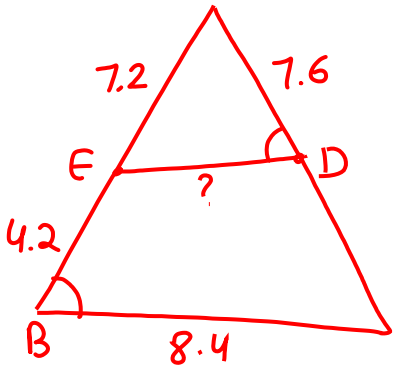
CGL 2019 T1



(3) In ΔPQR, PQ= 24 cm and $\angle Q = 58°$. S and T are the points on side PQ and PR respectively, such that $\angle STR = 122°$. If $PS = 14\ cm$ and $PT = 12\ cm$, then the length of RT is :

a) 14.8 cm
b) 16 cm
c) 15 cm
d) 16.4 cm

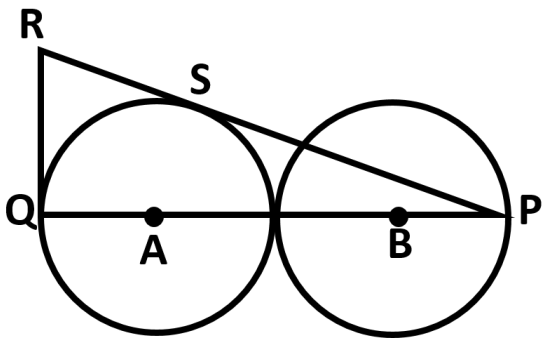
CGL 2019 T1



(4) In ΔABC, D and E are the points on sides AC and AB respectively such that $\angle ADE = \angle B$. If $AD = 7.6\ cm$, $AE = 7.2\ cm$, $BE = 4.2\ cm$ and $BC = 8.4$ cm then DE is equal to :

a) 6.3 cm
b) 5.6 cm
c) 7.4 cm
d) 5.6 cm

CHSL 2018

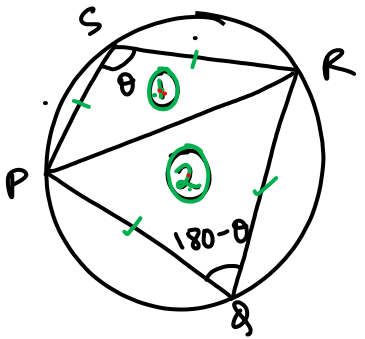


5 In the given figure, two identical circles of radius 4 cm touch each other, A and B are the centres of the two circles, if RQ is a tangent to the circle, then. What is the lenght (in cm) of RQ?

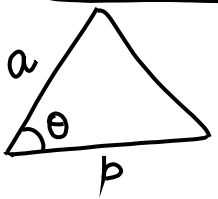
a) $3\sqrt{3}$ b) $2\sqrt{6}$

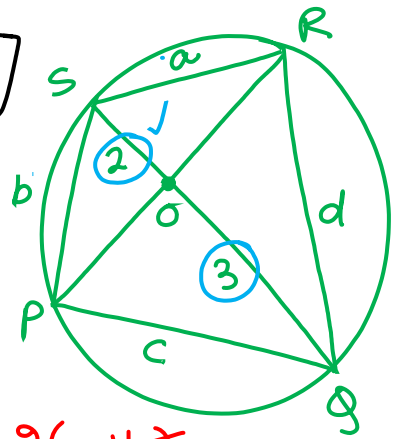
c) $4\sqrt{2}$ d) $\frac{16}{3}$

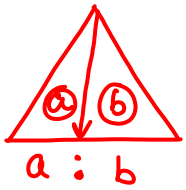
CGL 2017 T2



$$\text{ar} = \frac{1}{2}ab\sin\theta$$



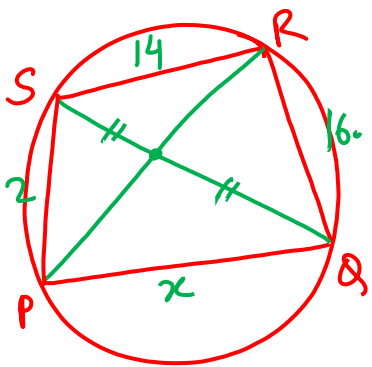




$$\frac{\text{ar } PSR}{\text{ar } PQR} = \frac{\frac{1}{2} \times PS \times SR \times \sin\theta}{\frac{1}{2} \times PQ \times QR \times \sin\theta} = \frac{2(a+b)}{3(a+b)}$$

$\sin\theta$ ↑ $\sin(180-\theta)$

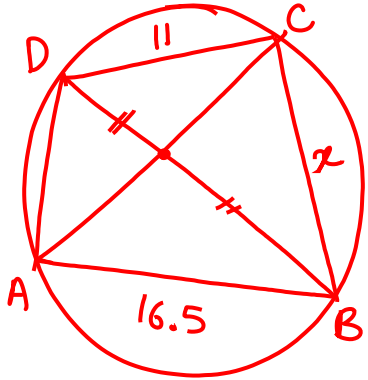
$$\frac{\text{ar } PSR}{\text{ar } PQR} = \frac{ab}{cd} = \frac{2}{3}$$



6 $PQRS$ is a cyclic quadrilateral in which $PQ = x\ cm, QR = 16.8\ cm, RS = 14$ cm, $PS = 25.2\ cm$ and PR bisects QS. What is the value of x?

a) 24 b) 21
c) 18 d) 28

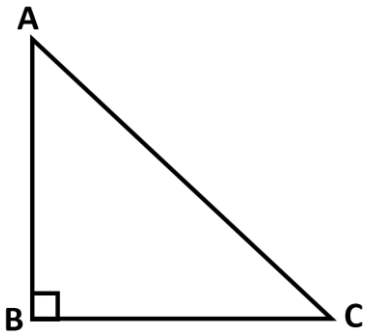
CGL 2019 T1



7 $ABCD$ is cyclic quadrilateral in which $AB = 16.5\ cm, BC = x\ cm,$ $CD = 11\ cm,\ AD = 19.8$ cm and BD is bisected by AC at O. What is the value of x?

a) 12.4 cm
b) 13.8 cm
c) 13.2 cm
d) 12.8 cm

CGL 2019 T1

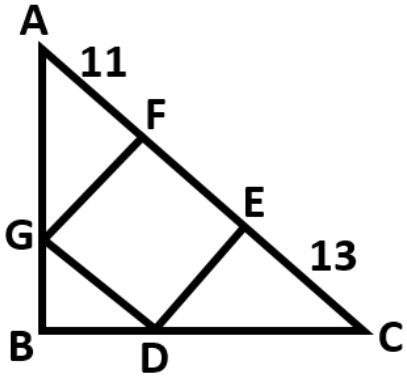


8 In the given figure, in a right angle triangle $ABC, AB = 12cm$ and $AC = 15\,cm$. A square is inscribed in the triangle. One of the vertices of square coincides with the vertex of triangle. What is the maximum possible area of the square?

a) $\frac{1296}{49}$ b) 25

c) $\frac{1225}{36}$ d) $\frac{1225}{64}$

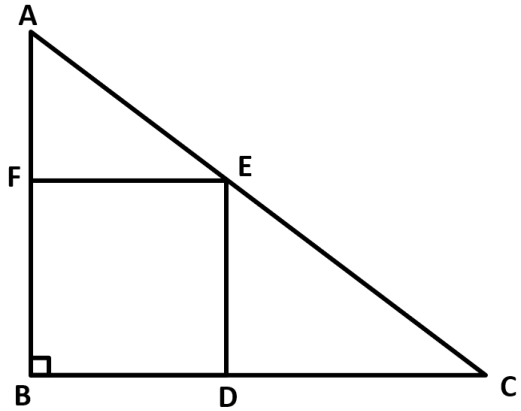
CGL 2017 T-2



(9) In a right angle triangle ABC, (right angled at B) there are two point on side AC such that $AF = 11\ cm$ and $EC = 13\ cm$. There are two points G and D on sides AB and BC respectively. Find the area of square $DEFG$.

a) 121 b) 169
c) 143 d) Can't say

SA



10 In the adjoining figure ABC is a right angled triangle, BDEF is a square, $AE = 15\ cm$ and $AC = 35\ cm$. What is the area of square BDEF?

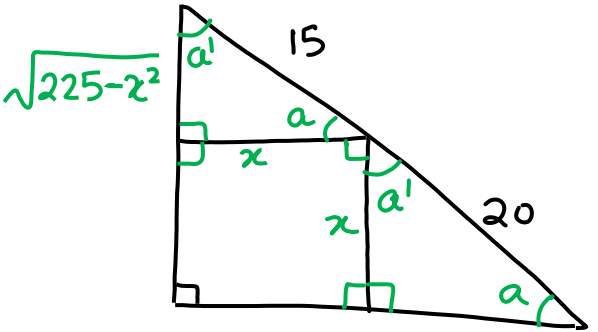
a) $169\ cm$ b) $144\ cm$

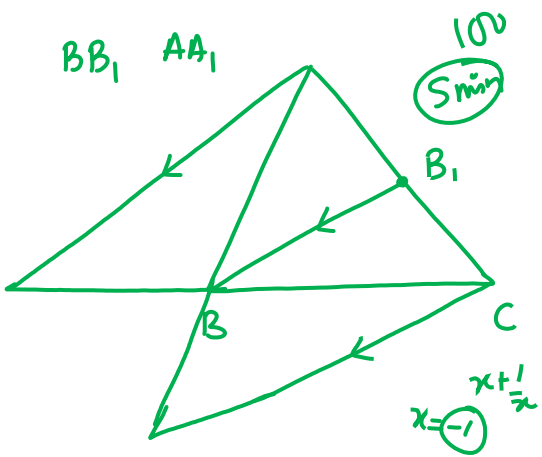
c) $225\ cm^2$ d) $100\ cm^2$

$$\frac{\sqrt{225-x^2}}{x} = \frac{\cancel{15}\ 3}{\cancel{20}\ 4}$$

$$16 \times 225 - 16x^2 = 9x^2$$

$$\Rightarrow 16 \times \overset{9}{\cancel{225}} = \cancel{25}x^2$$





(11) B_1 is a point on the side AC of $\Delta\ ABC$ and B_1B is joined. A line is drawn through A parallel to B_1B meeting CB at A_1 and another line is drawn through C parallel to B B meeting AB produced at C . Then

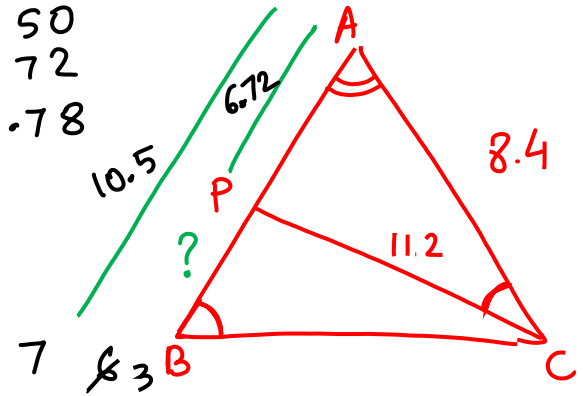
a) $\frac{1}{CC_1} - \frac{1}{AA_1} = \frac{1}{BB_1}$

b) $\frac{1}{CC_1} + \frac{1}{AA_1} = \frac{1}{BB_1}$

c) $\frac{1}{BB_1} - \frac{1}{AA_1} = \frac{2}{CC_1}$

d) $\frac{1}{AA_1} - \frac{1}{CC_1} = \frac{2}{BB_1}$

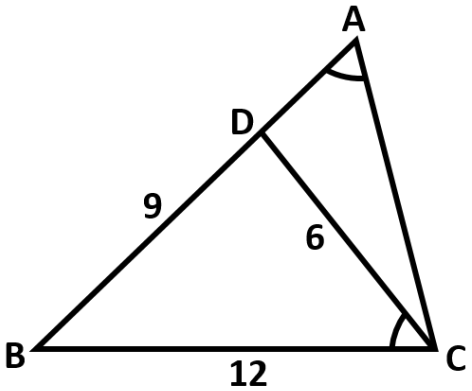
CGL 2016 T-2



(12) In ΔABC, $AC = 8.4\ cm$ and $BC = 14\ cm$. P is a point on AB such that $CP = 11.2\ cm$ and $\angle ACP = \angle B$. What is the length (in cm) of BP?

a) 3.78 b) 4.12
c) 3.6 d) 2.8

CGL 2019 T1



13 **Consider the triangle ABC shown in the following figure where BC = 12cm, DB = 9 cm, CD = 6 cm and $\angle BCD = \angle BAC$. What is the ratio of the perimeter of the triangle ADC to that of the triangle BDC?**

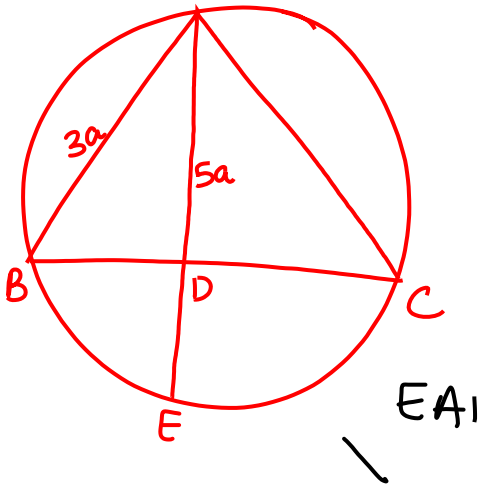
a) $\frac{7}{9}$ b) $\frac{8}{9}$ c) $\frac{6}{9}$ d) $\frac{5}{9}$

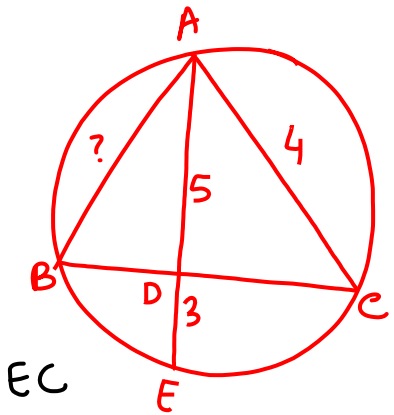
CAT 2005

(14) In ΔABC, the angle bisector of $\angle BAC$ cut BC at D and meet the circumcircle of ΔABC at E. If $AB: AD = 3:5$ *then find* $AE:AC$?

a) $2:3$ b) $3:5$

c) $5:3$ d) $3:2$

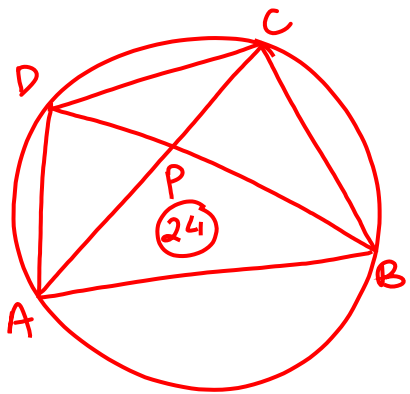




15 In a triangle ABC, angle bisector of $\angle BAC$ cut the side BC at D and meet the circumcircle of ΔABC at E. if $AC = 4\ cm, AD = 5\ cm, DE = 3\ cm$. Find the length of AB.

a) 10 b) 4 c) 15 d) 8

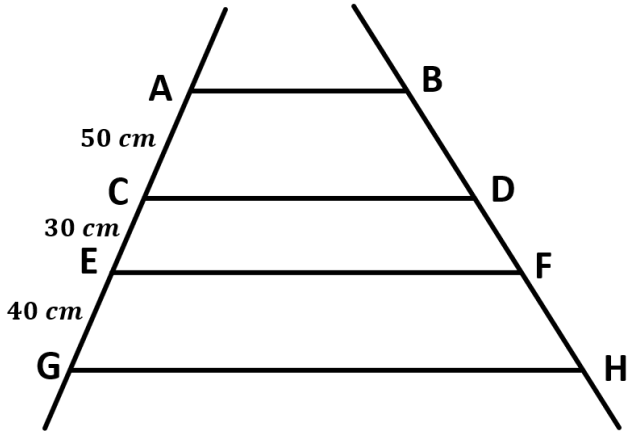
SA



16 The diagonals of a cyclic quadrilateral $ABCD$ intersect at P and the area of the triangle APB is 24 square cm. If $AB = 8cm$ and $CD = 5cm$, then what is the area of the triangle CPD?

a) 24 square cm
b) 15 square cm
c) 12.5 square cm
d) 9.375 square cm

CDS 2017 II



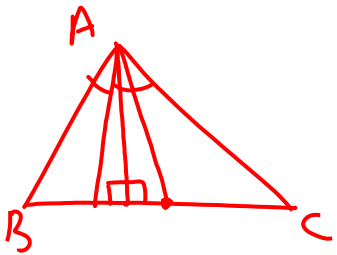
17 In the given figure $AB \parallel CD \parallel EF \parallel GH$ and $BH = 100\ cm$. Find the value of DF:

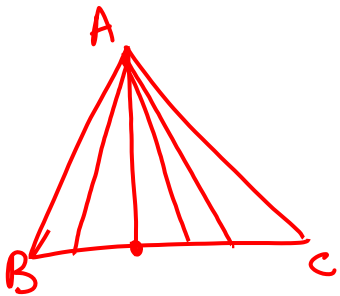
a) 26 cm b) 40cm

c) 25cm d) 24cm

Important cevians ?

1. Median
2. Altitude
3. Angle bisector
4. Perpendicular bisector

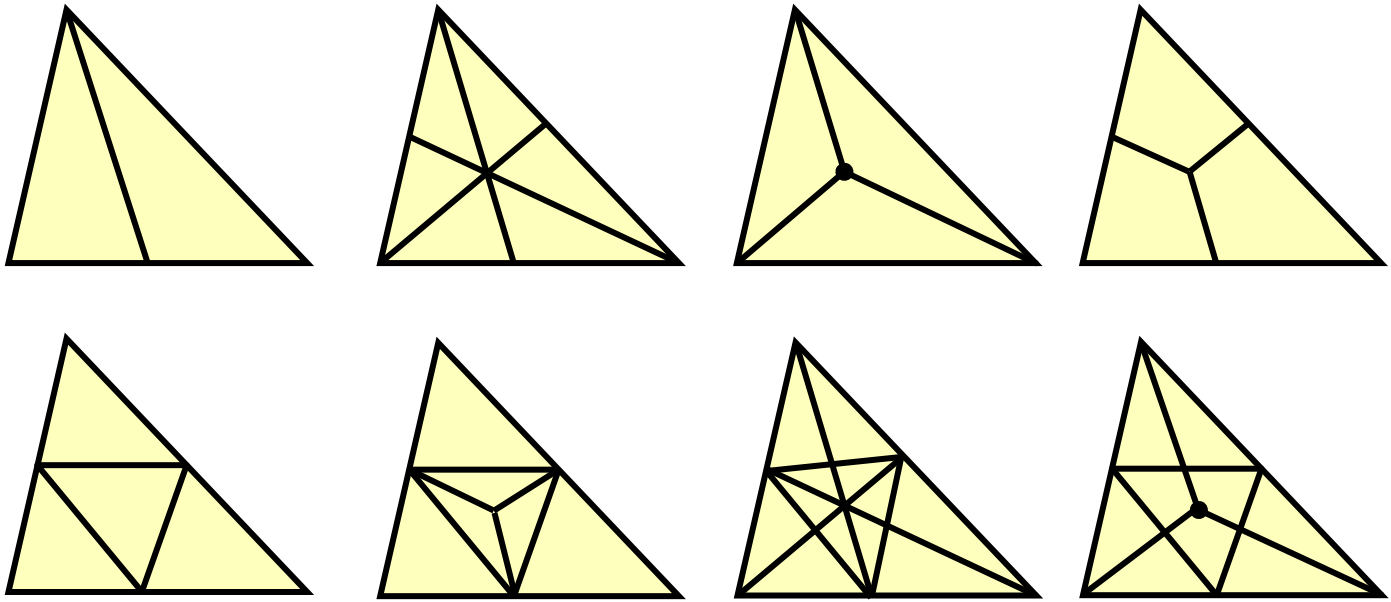




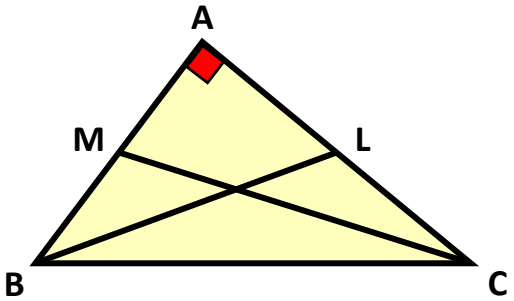
Median (

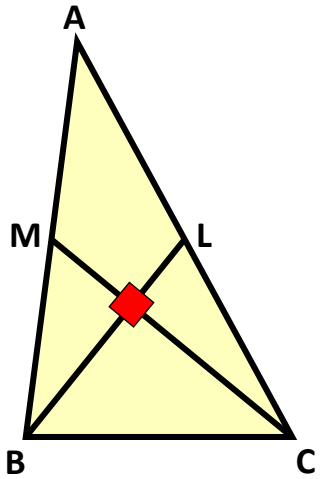
1. **Property:**
2. **Intersecting point:**
3. **Ratio property:**
4. **Location in different triangles:**

Area division by Medians (

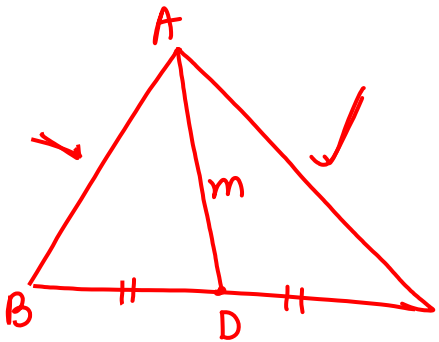


Median special cases:



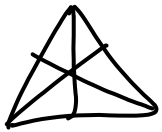


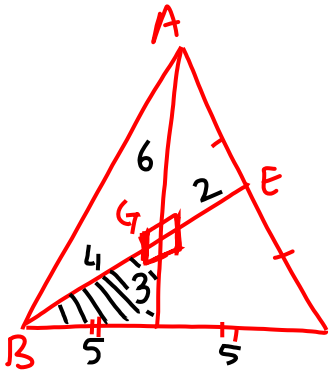
Apollonius theorem ( ):



Some standard results:

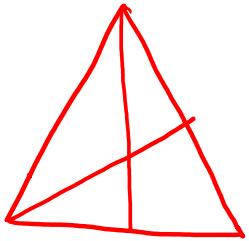
1. Sum of lengths of all medians < sum of lengths of all sides
2. Area of triangle formed by sides is $= \frac{4}{3} \times$ Area of triangle formed by its medians
3. Sum of squares of sides $= \frac{4}{3} \times$ Sum of squares of medians
4. Sum of sides $< \frac{4}{3} \times$ Sum of medians





(12) Two medians AD and BE of $\Delta\ ABC$ intersect at G at right angle. If $AD = 9\ cm$ and $BE = 6\ cm$, then the length of BC, in cm, is

a) 10 b) 6
c) 5 d) 3



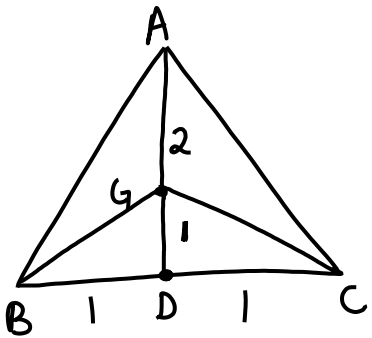
19 G is the centroid of ΔPQR. The medians PS and QT intersect at right angles. If the length of PS and QT are $60cm$ and $63\ cm$ respectively then the length of PQ?

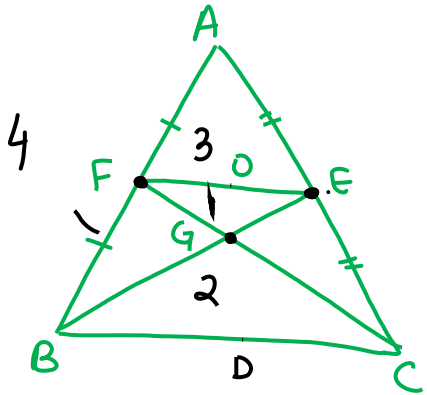
a) 48 b) 58
c) 62 d) 64

20 If G be the centroid of ΔABC such that $AG = BC$, then the magnitude of $\angle BGC$ is

a) 60 b) 90

c) 120 d) 135

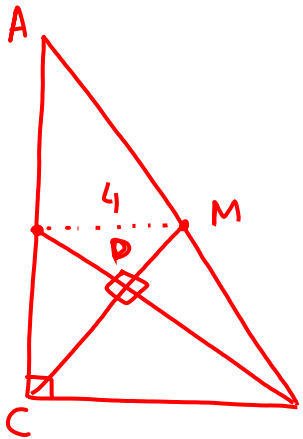




21 If BE and CF are two median of a ΔABC *and* G is there intersection point. Similarly O is intersecting point of EF and AG. Find $AO : OG$.

a) $1:1$ b) $1:2$

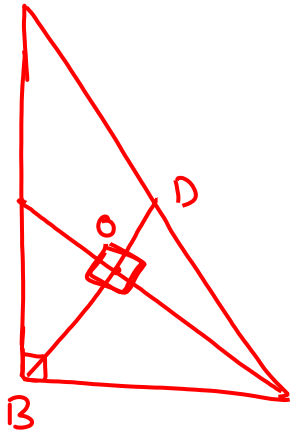
c) $2:1$ d) $3:1$



(22) In ΔABC, $\angle C = 90°$. M and N are the mid-points of sides AB and AC, respectively. CM and BN intersect each other at D and $\angle BDC = 90°$. If $MN = 4\ cm$, then the length of BN is:

a) $6\sqrt{3}\ cm$ b) $6\sqrt{6}\ cm$

c) $4\sqrt{6}\ cm$ d) $4\sqrt{3}\ cm$



(23) In a right angle triangle two medians intersects each other at $90°$. Find the ratio of all the three sides of the triangle?

a) $3:4:5$ b) $\sqrt{2}:\sqrt{3}:\sqrt{5}$

c) $1:\sqrt{2}:\sqrt{3}$ d) $\sqrt{3}:2:\sqrt{7}$